Nicolas Waehner

Buenos Aires, Argentina
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Education

University of Buenos Aires 2026 – Present
PhD in AI applied to Neuroscience. *Advisor: Enzo Tagliazucchi*

University of Buenos Aires 2019 – 2025
Licenciate in Physics (6 years degree program, 5072 hours). *GPA (scale 0–10): 9.43*

Research Experience

Summer research intern - Center for Astrophysics, Harvard & Smithsonian 2026
Developing SBI methods to detect biosignatures in exoplanet atmospheric spectra. *Advisor: Cecilia Garraffo*

Exoplanet detection using Neural Networks 2024 – 2025
Implementation of attention mechanisms for interpretability analysis of periodograms. *Advisor: Dr. Rodrigo F. Díaz*
Undergraduate Thesis Project. [GitHub] [Thesis]

Network Analysis of Argentine Artists 2022
Analysis of collaboration networks among Argentine artists using Spotify data obtained via web scraping. [GitHub] *Instructor: Dr. Ariel Chernomoretz*

Relevant Coursework

- Statistical Mechanics
- Machine Learning
- Modeling of Complex Systems
- Statistics for Experimental Physics
- Quantum Mechanics
- Linear Algebra
- Classical Mechanics
- Differential and Integral Calculus
- Complex Analysis

Achievements

Research Stipend – 10-week research program at the CfA, Harvard & Smithsonian, 2026.
Hackathon Winner – NLP-based investigation project, Y Hat Hackathon, 2025. [Poster]
Best Poster Award – Scientific Symposium on AI and Applications (SCIAA), UdeSA, 2025.
Honourable Mention Poster Award – First Spring School on Deep Learning, 2025.
Honourable Mention Project Award – Simulation-based Inference for Parameter Estimation in Neural Spiking Models, LASCON X, 2026.
Students' Achievements – Trainer of the Argentine delegation to the Latin American Olympiad of Astronomy and Astronautics (OLAA) 2025: 1 gold medal, 2 silver, and 1 bronze.
Professor Academic Distinction – Awarded by the Rector of the University of Buenos Aires (UBA), 2025.
Gold Medal – National Astronomy Olympiad, 2019.

Teaching Experience

Teaching Assistant, Faculty of Exact and Natural Sciences – UBA Mar 2024 – Mar 2026
Physics I (Mechanics), Physics III (Electromagnetism), and Laboratory V. Teaching practical classes, Q&A sessions, grading assignments and exams.

Astronomy Teacher, Colegio Nacional de Buenos Aires Sept 2023 – Mar 2026
Instructor of a yearly elective astronomy course. Trainer of the Argentine delegation to the Latin American Olympiad of Astronomy and Astronautics 2025.

Schools and Conferences

AstroAI Workshop — Center for Astrophysics — Harvard & Smithsonian — June 2026
LASCON X — Latin American School on Computational Neuroscience, São Paulo — January 2026
4th School on Data Science and Machine Learning — ICTP-SAIFR, São Paulo — November 2025
Spring School on Deep Learning — UBA — October 2025
SCIAA — Scientific Symposium on AI and Applications — UdeSA — November 2024 & September 2025

Conference Presentations

Nicolás Waehner, Luis Agustín Nieto, Rodrigo F. Díaz. *Machine learning for exoplanet detection through the radial velocity method*. [Poster]

- **Poster presented at:** Scientific Symposium on Artificial Intelligence and Applications, Deep Learning Spring School, 4th School on Data Science and Machine Learning, ICTP-SAIFR and AstroAI Workshop at Center for Astrophysics.

Nicolás Waehner, Lautaro Cabral, Oscar E. Martinez, Micaela Toscani. *Improvement of a miniature fluorescence microscope for applications in neuroscience*.

- **Poster presented at:** Annual Meeting of the Argentine Physics Association 2024.

Science Outreach

Teaching Assistant at: Python Workshop (FIFA 2022).
Presenter at: "La Noche de los Museos" organized by the University of Buenos Aires (2019, 2023, 2024); Presenter at the "Physics Week" organized by the University of Buenos Aires (2024).
Organization and coordination: Co-organizer and grader at the Astronomical Challenges Conference (2025); Writer and grader at the Metropolitan Physics Olympiad (2025).

Languages

- **Spanish**: Native.
- **English**: Proficient (C1).